Per-LR lookaheads and the joint 5-interval search: live results

Updated 2026-10-23 16:35 PDT (23:35 UTC). BS 2000, CIFAR-10 airbench, score = TTA test accuracy, one training seed (0). Jobs: 19 done, 8 running, 3 pending, 0 failed.

A. Capacity of the 5-constant-interval conv schedule: joint neighbourhood search

Setup. Conv LR only (the other groups on the baseline); conv's specified schedule is 5 constant intervals on the x0.6 grid, and the transform multiplies conv's applied LR. One search seed (0). Each iteration trains all 3^5 - 1 = 242 joint neighbours (every interval -1, 0 or +1 grid step) as full runs, moves to the best if it beats the current schedule, and repeats until none does. Start: the oracle (baseline / multiplier) reduced to its 5 interval means, snapped to the grid. The result is the winner's search score (no replay); baseline and do-nothing are single runs on the same seed.

Table A. Joint search per transform. Baseline (no transform, seed 0): 93.92. Iterations = moves made so far; best evaluated = the best full run so far (the schedule the search moves to next); a running search's best is final only once it converges.

Transform	do nothing	start (score)	best evaluated (score)	Δ vs baseline	iterations	runs	status
no transform	93.92	0.22, 0.13, 0.13, 0.078, 0.028 (93.53)	0.078, 0.13, 0.078, 0.047, 0.01 (94.09)	+0.17	3	682	converged
ramp x0.25->x4	93.31	0.6, 0.22, 0.13, 0.047, 0.01 (93.13)	0.36, 0.36, 0.078, 0.047, 0.006 (93.88)	-0.04	1	438	converged
reverse ramp x4->x0.25	93.47	0.078, 0.078, 0.13, 0.13, 0.078 (93.61)	0.078, 0.078, 0.047, 0.047, 0.017 (94.17)	+0.25	3	711	converged
wave x2^sin	93.68	0.13, 0.078, 0.13, 0.13, 0.047 (93.19)	0.078, 0.078, 0.13, 0.13, 0.017 (93.93)	+0.01	2	531	converged
inverted wave	93.24	0.36, 0.36, 0.13, 0.047, 0.017 (93.73)	0.36, 0.36, 0.078, 0.028, 0.01 (94.03)	+0.11	1	414	converged
steep ramp x1/8->x8	92.47	1, 0.36, 0.13, 0.028, 0.006 (93.07)	1, 0.22, 0.078, 0.01, 0.0013 (93.84)	-0.08	3	744	converged

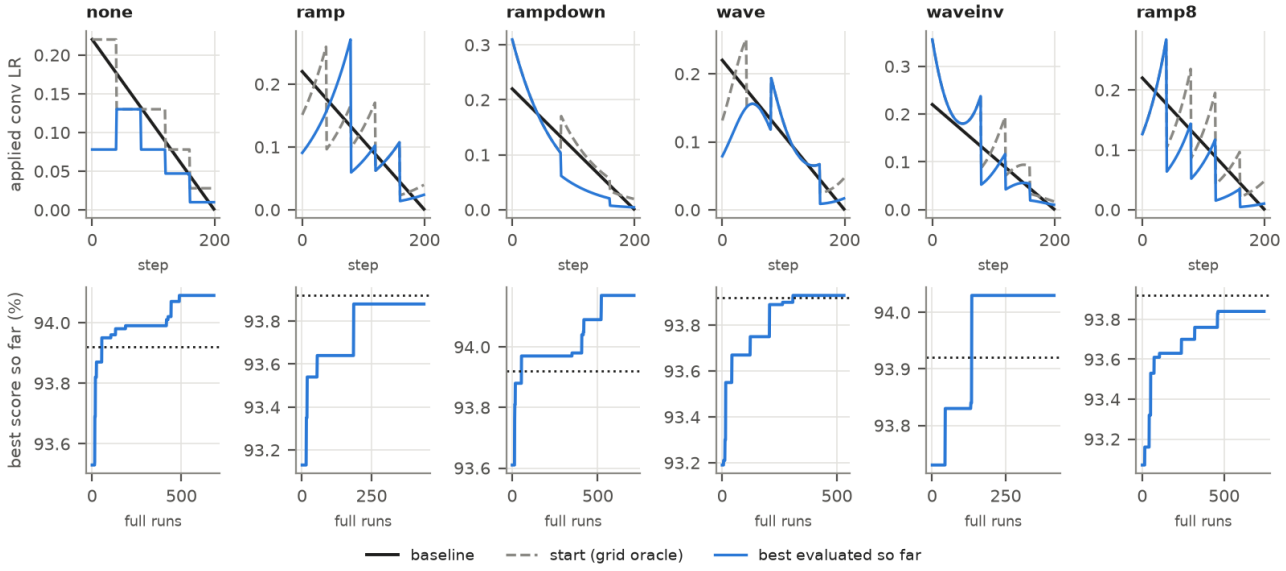


Figure A. Top: applied conv LR of the start and the best evaluated schedule against the baseline. Bottom: best score among the full runs so far (dotted: baseline, seed 0).

B. Which one-interval lookahead ranks interval candidates like the truth?

Setup. The plan is the baseline run itself, split into 5 intervals of 40 steps. Under a transform, the multiplier applies to all four LR groups, and the plan is the oracle (each group's baseline divided by the multiplier), so the plan's applied LRs are the baseline's. For a main LR and interval k, the plan is followed up to k; there the main LR is replaced by a constant candidate c on the x0.6 grid (the grid value nearest the plan's mean over the interval, and 1 and 2 grid steps either side; whitening bias, whose plan is 0 from step 75, uses its step-0 plan value), trained for one interval from the same state, and scored by one more interval of: **linear to 0, main LR only** (c to 0, the others on the plan); **linear to 0, all LRs**; **searched m, main LR only** (constant m x c); **searched m main + tied m others** (the other three at m' x their plan mean over interval k, one m' for all three); **searched m for each LR**. The **truth** is the final score when the run continues on the plan. Each m is searched on the 0.6^k grid without bounds, from 0.22, one multiplier at a time, moving while a neighbour at k±1 or k±2 is better; the result beats all four neighbours. A lookahead decides like the truth when its curve peaks at the same candidate. One seed: a single-run score has an SD of about 0.12 pp, so differences below ~0.15 pp between candidates are within noise.

B.1 no transform

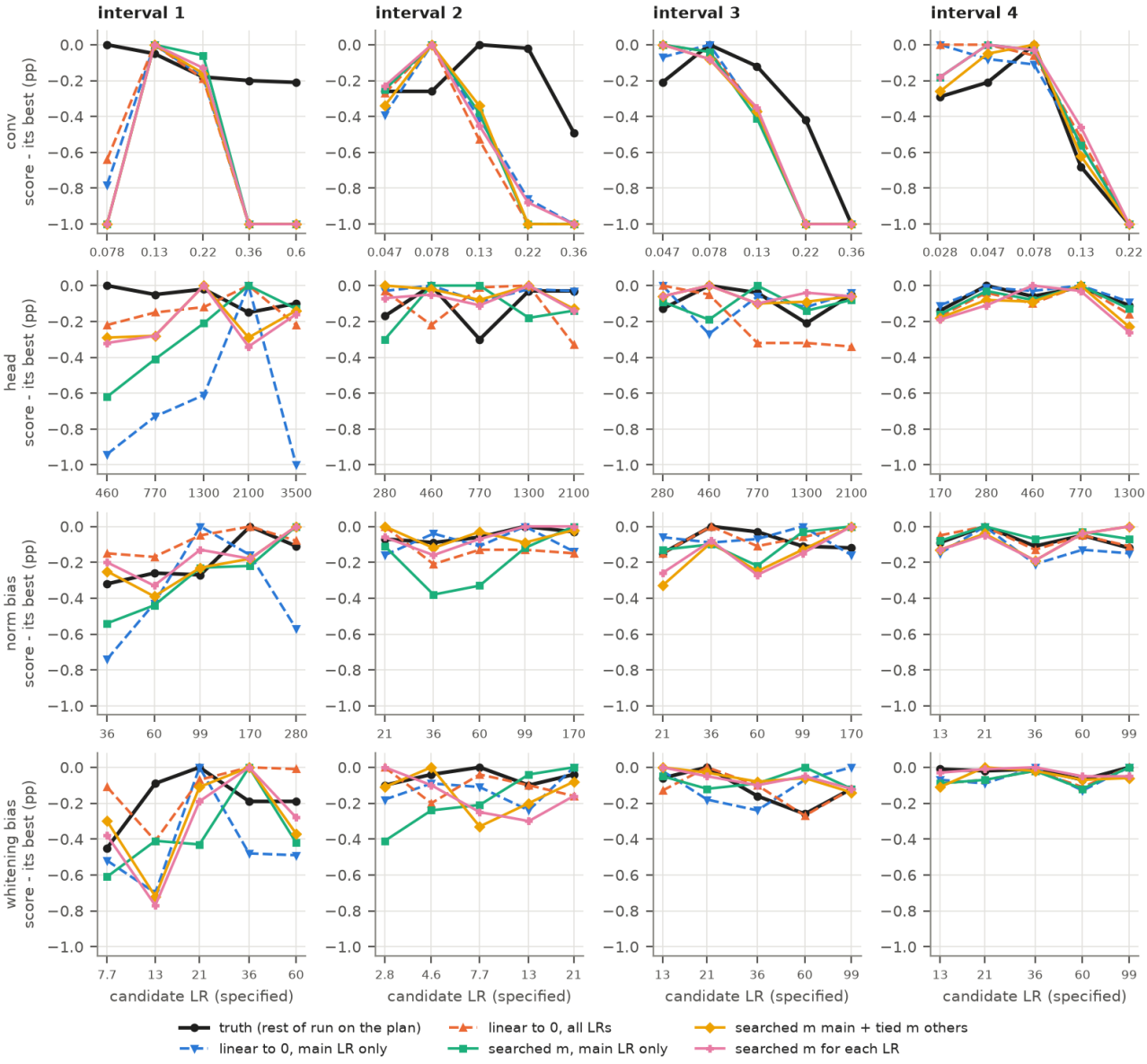


Figure B1. no transform: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table B1. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.078	0.13 (0.05)	0.13 (0.05)	0.13 (0.05)	0.13 (0.05)	0.13 (0.05)
conv, 2	0.13	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)
conv, 3	0.078	**0.078 (0.00)**	0.047 (0.21)	0.047 (0.21)	0.047 (0.21)	0.047 (0.21)
conv, 4	0.078	0.028 (0.29)	0.028 (0.29)	0.047 (0.21)	**0.078 (0.00)**	0.047 (0.21)
head, 1	460	2100 (0.15)	2100 (0.15)	2100 (0.15)	1300 (0.02)	1300 (0.02)
head, 2	460	**460 (0.00)**	1300 (0.03)	**460 (0.00)**	280 (0.17)	1300 (0.03)
head, 3	460	280 (0.13)	280 (0.13)	770 (0.04)	**460 (0.00)**	**460 (0.00)**
head, 4	280	770 (0.01)	770 (0.01)	770 (0.01)	770 (0.01)	460 (0.06)
norm bias, 1	170	99 (0.27)	**170 (0.00)**	280 (0.11)	280 (0.11)	280 (0.11)
norm bias, 2	99	**99 (0.00)**	21 (0.07)	170 (0.03)	21 (0.07)	**99 (0.00)**
norm bias, 3	36	99 (0.11)	**36 (0.00)**	170 (0.12)	170 (0.12)	170 (0.12)
norm bias, 4	21	**21 (0.00)**	**21 (0.00)**	**21 (0.00)**	99 (0.12)	99 (0.12)
whitening bias, 1	21	**21 (0.00)**	36 (0.19)	36 (0.19)	36 (0.19)	36 (0.19)
whitening bias, 2	7.7	21 (0.04)	2.8 (0.10)	21 (0.04)	4.6 (0.04)	2.8 (0.10)
whitening bias, 3	21	99 (0.12)	**21 (0.00)**	60 (0.26)	13 (0.06)	13 (0.06)
whitening bias, 4	99	36 (0.01)	**99 (0.00)**	**99 (0.00)**	21 (0.02)	36 (0.01)

Table B1m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 9/20; else 0.13: 6, 0.36: 3, 0.078: 1	0.22 in 8/20; else 0.13: 6, 0.36: 5, 0.078: 1	0.22 in 3/20; else 0.6: 5, 1.7: 3, 0.078: 3	0.22 in 7/20; else 0.13: 7, 0.36: 5, 0.078: 1
head	0.22 in 2/20; else 0.36: 5, 0.6: 4, 0.078: 4, 0.13: 2	0.22 in 3/20; else 0.13: 4, 0.36: 4, 0.078: 3, 0.6: 3	0.22 in 16/20; else 0.13: 4	0.22 in 4/20; else 0.36: 5, 0.13: 4, 0.6: 4
norm bias	0.22 in 1/20; else 0.36: 6, 1: 3, 0.078: 3, 0.13: 2	0.22 in 5/20; else 0.6: 4, 0.13: 3, 0.36: 3	0.22 in 19/20; else 0.13: 1	0.22 in 5/20; else 0.6: 6, 0.13: 3, 0.36: 3
whitening bias	0.22 in 5/20; else 0.6: 4, 0.13: 3, 0.078: 2	0.22 in 4/20; else 0.6: 6, 0.36: 4, 1.7: 2	0.22 in 17/20; else 0.13: 3	0.22 in 6/20; else 0.36: 6, 0.6: 5, 0.078: 1

B.2 ramp x0.25->x4

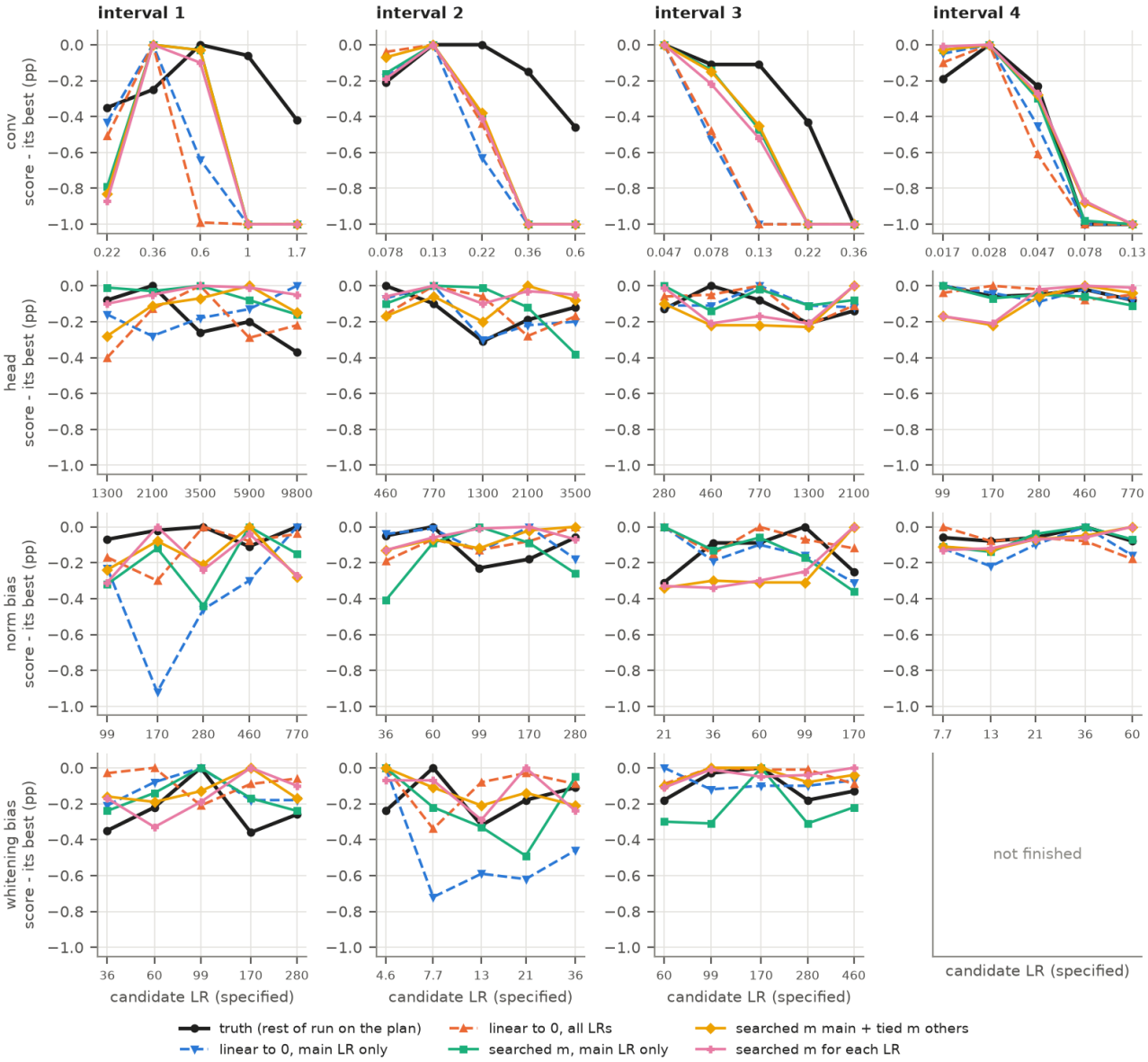


Figure B2. ramp x0.25->x4: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table B2. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.6	0.36 (0.25)	0.36 (0.25)	0.36 (0.25)	0.36 (0.25)	0.36 (0.25)
conv, 2	0.13	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**
conv, 3	0.047	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**
conv, 4	0.028	**0.028 (0.00)**	**0.028 (0.00)**	**0.028 (0.00)**	**0.028 (0.00)**	**0.028 (0.00)**
head, 1	2100	9800 (0.37)	3500 (0.26)	3500 (0.26)	5900 (0.20)	3500 (0.26)
head, 2	460	770 (0.10)	770 (0.10)	770 (0.10)	2100 (0.19)	770 (0.10)
head, 3	460	770 (0.08)	770 (0.08)	280 (0.13)	2100 (0.14)	2100 (0.14)
head, 4	99	**99 (0.00)**	170 (0.06)	**99 (0.00)**	460 (0.02)	460 (0.02)
norm bias, 1	280	770 (0.00)	**280 (0.00)**	460 (0.11)	460 (0.11)	170 (0.02)
norm bias, 2	60	170 (0.18)	280 (0.06)	99 (0.23)	280 (0.06)	170 (0.18)
norm bias, 3	99	21 (0.31)	21 (0.31)	21 (0.31)	170 (0.25)	170 (0.25)
norm bias, 4	36	**36 (0.00)**	7.7 (0.06)	**36 (0.00)**	60 (0.08)	60 (0.08)
whitening bias, 1	99	**99 (0.00)**	60 (0.22)	**99 (0.00)**	170 (0.36)	170 (0.36)
whitening bias, 2	7.7	4.6 (0.24)	4.6 (0.24)	4.6 (0.24)	4.6 (0.24)	21 (0.18)
whitening bias, 3	170	60 (0.18)	99 (0.03)	**170 (0.00)**	99 (0.03)	460 (0.13)

Table B2m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 2/20; else 0.13: 9, 0.078: 8, 0.047: 1	0.22 in 3/20; else 0.078: 10, 0.13: 6, 0.047: 1	0.22 in 6/20; else 0.6: 5, 0.36: 5, 1: 1	0.22 in 3/20; else 0.078: 11, 0.13: 6
head	0.22 in 6/20; else 0.078: 5, 1: 3, 0.6: 2	0.22 in 0/20; else 0.36: 7, 1.7: 3, 0.13: 3, 0.078: 3	0.22 in 0/20; else 0.078: 17, 0.13: 2, 0.047: 1	0.22 in 3/20; else 0.078: 4, 0.36: 3, 0.6: 3
norm bias	0.22 in 6/20; else 0.36: 4, 0.6: 3, 0.078: 3	0.22 in 7/20; else 0.078: 4, 0.13: 3, 0.6: 2	0.22 in 0/20; else 0.078: 17, 0.13: 2, 0.047: 1	0.22 in 4/20; else 0.36: 4, 0.13: 4, 0.078: 3
whitening bias	0.22 in 4/15; else 0.13: 4, 1: 2, 0.078: 2	0.22 in 1/15; else 0.6: 5, 0.13: 3, 0.078: 2, 1.7: 1	0.22 in 0/15; else 0.078: 13, 0.13: 2	0.22 in 4/15; else 0.078: 4, 0.36: 3, 0.6: 2

B.3 reverse ramp x4->x0.25

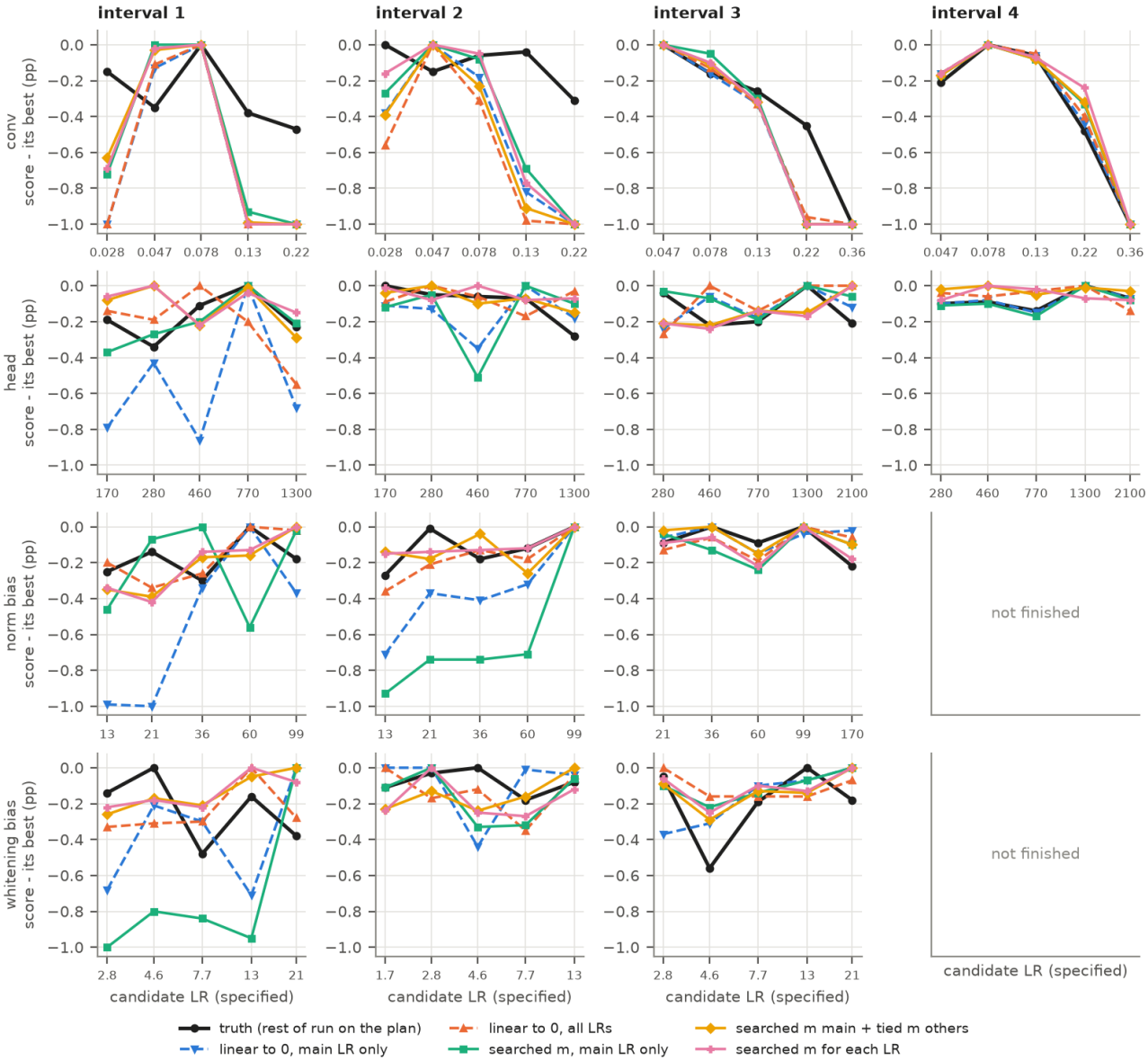


Figure B3. reverse ramp x4->x0.25: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table B3. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.078	**0.078 (0.00)**	**0.078 (0.00)**	0.047 (0.35)	**0.078 (0.00)**	**0.078 (0.00)**
conv, 2	0.028	0.047 (0.15)	0.047 (0.15)	0.047 (0.15)	0.047 (0.15)	0.047 (0.15)
conv, 3	0.047	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**
conv, 4	0.078	**0.078 (0.00)**	**0.078 (0.00)**	**0.078 (0.00)**	**0.078 (0.00)**	**0.078 (0.00)**
head, 1	770	**770 (0.00)**	460 (0.11)	**770 (0.00)**	280 (0.34)	280 (0.34)
head, 2	170	770 (0.07)	280 (0.05)	770 (0.07)	280 (0.05)	460 (0.06)
head, 3	1300	**1300 (0.00)**	460 (0.22)	**1300 (0.00)**	2100 (0.21)	2100 (0.21)
head, 4	1300	**1300 (0.00)**	**1300 (0.00)**	**1300 (0.00)**	460 (0.09)	460 (0.09)
norm bias, 1	60	**60 (0.00)**	**60 (0.00)**	36 (0.30)	99 (0.18)	99 (0.18)
norm bias, 2	99	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**
norm bias, 3	36	**36 (0.00)**	99 (0.00)	99 (0.00)	**36 (0.00)**	99 (0.00)
whitening bias, 1	4.6	21 (0.38)	13 (0.16)	21 (0.38)	21 (0.38)	13 (0.16)
whitening bias, 2	4.6	1.7 (0.11)	1.7 (0.11)	2.8 (0.03)	13 (0.08)	2.8 (0.03)
whitening bias, 3	13	21 (0.18)	2.8 (0.05)	21 (0.18)	21 (0.18)	21 (0.18)

Table B3m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 5/20; else 0.36: 9, 0.6: 4, 1: 2	0.22 in 6/20; else 0.36: 8, 0.6: 5, 1: 1	0.22 in 6/20; else 0.13: 4, 0.36: 3, 4.6: 2	0.22 in 5/20; else 0.36: 8, 0.6: 6, 1: 1
head	0.22 in 3/20; else 0.13: 5, 0.36: 4, 0.6: 3	0.22 in 3/20; else 0.36: 4, 2.8: 3, 0.6: 2	0.22 in 2/20; else 0.36: 17, 0.6: 1	0.22 in 2/20; else 0.6: 5, 0.078: 4, 0.36: 4, 1: 3
norm bias	0.22 in 3/15; else 0.36: 6, 0.078: 2, 7.7: 1	0.22 in 3/15; else 1.7: 3, 0.13: 3, 7.7: 2	0.22 in 1/15; else 0.36: 13, 0.6: 1	0.22 in 2/15; else 0.6: 4, 0.36: 3, 0.078: 3
whitening bias	0.22 in 1/15; else 0.6: 6, 0.13: 3, 0.078: 2, 0.028: 2	0.22 in 3/15; else 0.6: 3, 0.078: 3, 0.047: 1	0.22 in 0/15; else 0.36: 14, 0.6: 1	0.22 in 4/15; else 0.6: 5, 0.36: 3, 0.13: 2

B.4 wave x2^sin

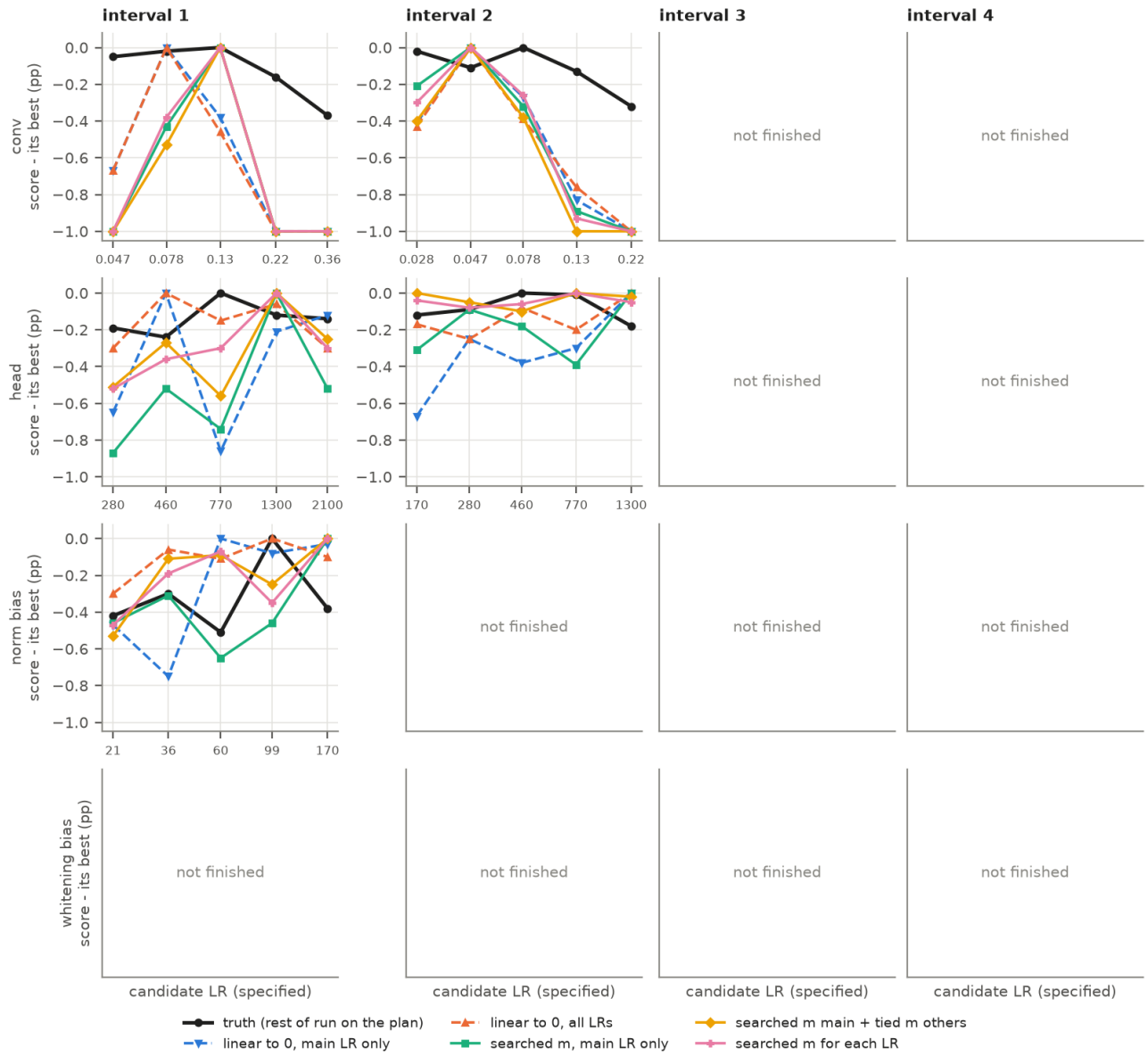


Figure B4. wave x2^sin: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table B4. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.13	0.078 (0.02)	0.078 (0.02)	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**
conv, 2	0.078	0.047 (0.11)	0.047 (0.11)	0.047 (0.11)	0.047 (0.11)	0.047 (0.11)
head, 1	770	460 (0.24)	460 (0.24)	1300 (0.12)	1300 (0.12)	1300 (0.12)
head, 2	460	1300 (0.18)	1300 (0.18)	1300 (0.18)	170 (0.12)	770 (0.01)
norm bias, 1	99	60 (0.51)	**99 (0.00)**	170 (0.38)	170 (0.38)	170 (0.38)

Table B4m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 3/10; else 0.36: 2, 0.6: 2, 0.13: 1	0.22 in 3/10; else 0.36: 3, 0.6: 2, 0.13: 1	0.22 in 0/10; else 0.36: 3, 0.6: 2, 2.8: 2, 1: 1	0.22 in 3/10; else 0.36: 3, 0.6: 2, 0.13: 1
head	0.22 in 4/10; else 0.36: 2, 0.078: 2, 0.6: 1	0.22 in 2/10; else 0.36: 5, 1: 1, 1.7: 1	0.22 in 3/10; else 0.36: 4, 0.13: 2, 0.6: 1	0.22 in 1/10; else 0.6: 3, 0.36: 3, 1.7: 1, 0.078: 1
norm bias	0.22 in 1/5; else 0.36: 2, 7.7: 1, 0.047: 1	0.22 in 0/5; else 0.6: 2, 1: 1, 1.7: 1, 0.36: 1	0.22 in 2/5; else 0.13: 3	0.22 in 1/5; else 1.7: 2, 0.6: 2
whitening bias	–	–	–	–

B.6 steep ramp x1/8->x8

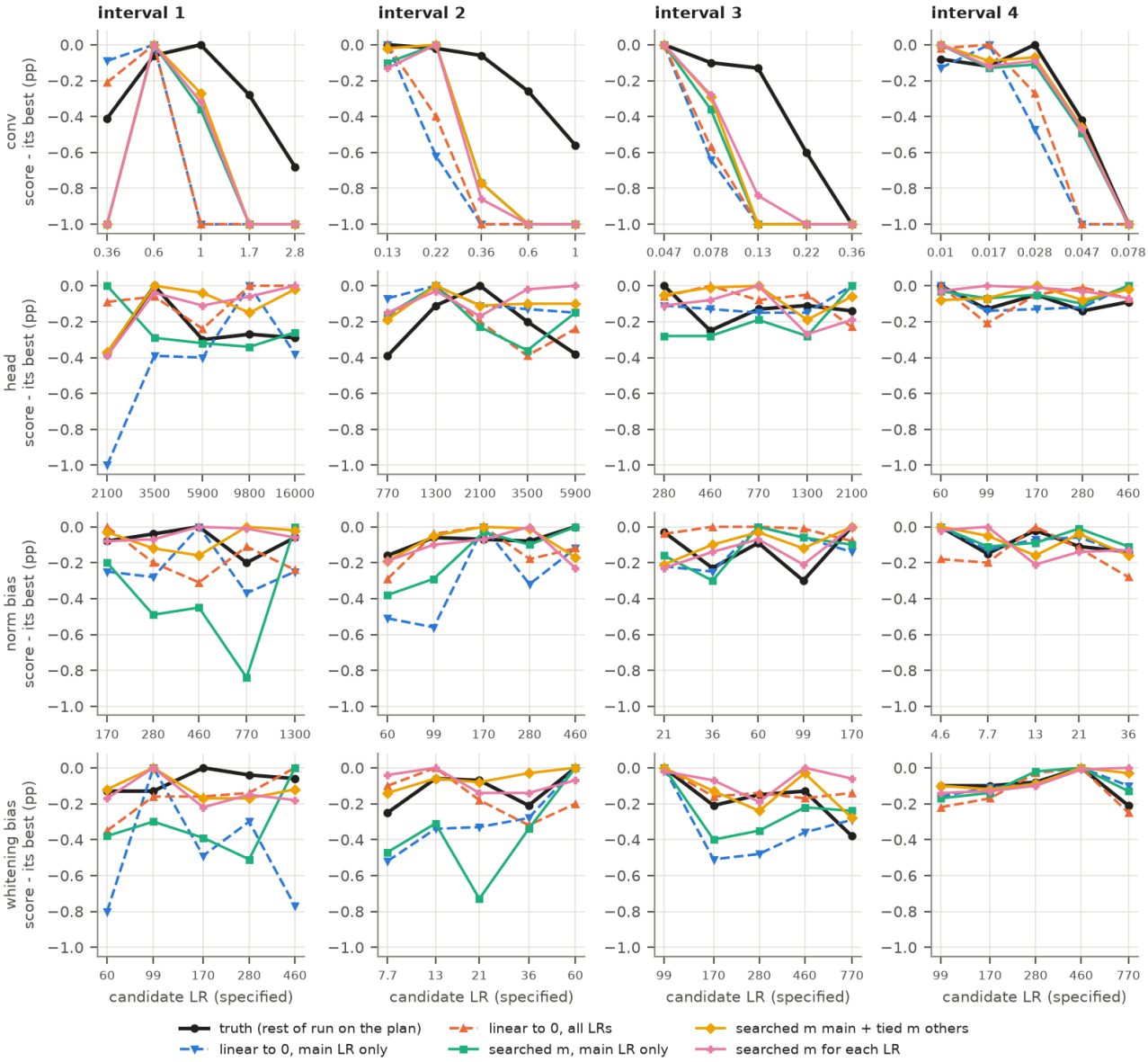


Figure B6. steep ramp x1/8->x8: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table B6. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	1	0.6 (0.06)	0.6 (0.06)	0.6 (0.06)	0.6 (0.06)	0.6 (0.06)
conv, 2	0.13	**0.13 (0.00)**	**0.13 (0.00)**	0.22 (0.02)	0.22 (0.02)	0.22 (0.02)
conv, 3	0.047	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**
conv, 4	0.028	0.017 (0.12)	0.017 (0.12)	0.01 (0.08)	0.01 (0.08)	0.01 (0.08)
head, 1	3500	9800 (0.27)	9800 (0.27)	2100 (0.38)	**3500 (0.00)**	16000 (0.29)
head, 2	2100	1300 (0.11)	1300 (0.11)	1300 (0.11)	1300 (0.11)	5900 (0.38)
head, 3	280	2100 (0.14)	460 (0.25)	2100 (0.14)	770 (0.13)	770 (0.13)
head, 4	60	**60 (0.00)**	**60 (0.00)**	460 (0.09)	170 (0.05)	99 (0.13)
norm bias, 1	460	**460 (0.00)**	170 (0.08)	1300 (0.06)	770 (0.20)	**460 (0.00)**
norm bias, 2	460	170 (0.07)	170 (0.07)	**460 (0.00)**	170 (0.07)	280 (0.08)
norm bias, 3	170	60 (0.09)	36 (0.23)	60 (0.09)	**170 (0.00)**	**170 (0.00)**
norm bias, 4	4.6	**4.6 (0.00)**	13 (0.02)	**4.6 (0.00)**	**4.6 (0.00)**	7.7 (0.15)
whitening bias, 1	170	99 (0.13)	460 (0.06)	460 (0.06)	99 (0.13)	99 (0.13)
whitening bias, 2	60	**60 (0.00)**	13 (0.06)	**60 (0.00)**	**60 (0.00)**	13 (0.06)
whitening bias, 3	99	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**	460 (0.13)
whitening bias, 4	460	**460 (0.00)**	**460 (0.00)**	**460 (0.00)**	**460 (0.00)**	770 (0.21)

Table B6m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 0/20; else 0.078: 11, 0.13: 5, 0.047: 3, 0.028: 1	0.22 in 0/20; else 0.078: 12, 0.13: 5, 0.047: 3	0.22 in 6/20; else 0.36: 4, 0.078: 3, 0.13: 3	0.22 in 0/20; else 0.078: 13, 0.13: 4, 0.047: 3
head	0.22 in 4/20; else 0.6: 6, 0.13: 4, 0.078: 2	0.22 in 1/20; else 0.13: 3, 0.047: 3, 0.36: 3, 0.078: 3	0.22 in 0/20; else 0.047: 14, 0.078: 5, 0.028: 1	0.22 in 3/20; else 0.078: 5, 0.6: 5, 0.36: 3
norm bias	0.22 in 6/20; else 0.6: 4, 0.028: 3, 0.078: 3	0.22 in 1/20; else 0.13: 7, 0.36: 4, 0.6: 3, 1: 2	0.22 in 0/20; else 0.047: 10, 0.078: 10	0.22 in 4/20; else 0.36: 6, 0.13: 5, 0.078: 2
whitening bias	0.22 in 4/20; else 0.13: 7, 0.078: 4, 0.36: 3	0.22 in 2/20; else 0.13: 6, 0.36: 5, 0.6: 2	0.22 in 0/20; else 0.047: 12, 0.078: 8	0.22 in 4/20; else 0.36: 4, 1.7: 3, 0.6: 3

Not finished yet (no complete interval cell): inverted wave.